

ENG290-2

Applied Cryptography for Embedded Systems

Secure LoRa Wireless Message Transmission System

Preliminary Design Review & Product Design Specification

Group Members

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Introduction

This document presents the preliminary design and product specification of a secure wireless communication system. The system allows two computers to exchange messages through STM32F042-based nodes and LoRa wireless communication. The design focuses on providing message confidentiality and integrity while maintaining a simple and low-cost embedded implementation.

The document defines the main system requirements, preliminary hardware and software architecture, technical risks, and project execution plan. The proposed system uses AES-128 encryption with a pre-shared key and a Message Authentication Code (MAC) to protect transmitted messages.

1 Executive Summary & Project Scope

1.1 Context

Wireless communication is vulnerable to eavesdropping and message modification because data is transmitted through an open radio channel. This project addresses this problem by developing a secure wireless communication system for exchanging messages between two computers through LoRa-enabled embedded devices.

The intended application is a low-cost secure messaging system in which confidentiality and message integrity are required. Each computer communicates with an STM32F042 microcontroller, which processes the data before it is transmitted over the LoRa wireless link.

1.2 High-Level Solution

The proposed system consists of two STM32F042-based communication nodes equipped with Ra-02 LoRa modules. Each computer provides the user interface through a command-line application and communicates with its STM32 through USB. The STM32 performs the required security processing and communicates with the LoRa module using SPI.

For confidentiality, application messages are encrypted using AES-128 with a pre-shared key. A Message Authentication Code (MAC) is used to verify the integrity and authenticity of received messages. The receiving node verifies the MAC before accepting the message and then decrypts valid data before forwarding it to the receiving computer.

1.3 Primary Scope

The following features are **in scope** for this semester:

- Secure communication between two computers through LoRa-enabled STM32F042 nodes.
- AES-128 encryption for message confidentiality.
- A pre-shared key for cryptographic operations.
- MAC-based message integrity and authentication.
- USB communication between each computer and its STM32F042 node.

- SPI communication between the STM32F042 and the Ra-02 LoRa module.
- A basic command-line application for entering and receiving messages.

The following features are **out of scope**:

- Public-key cryptography and digital signatures.
- Internet connectivity or cloud-based communication.
- Production-level key management or remote key distribution.

2 Product Design Specification (PDS)

The Product Design Specification defines the main functional, performance, interface, and design constraints of the proposed secure wireless message transmission system.

2.1 Functional Requirements

Table 1: Functional Requirements

ID	Requirement
FR-01	The system shall provide secure transmission of messages between two computers through LoRa-enabled STM32F042 nodes.
FR-02	The system shall encrypt transmitted messages using AES-128.
FR-03	The system shall use a pre-shared key for cryptographic operations.
FR-04	The system shall use a Message Authentication Code (MAC) to provide message integrity and authentication.

2.2 Performance Requirements

Table 2: Performance Requirements

ID	Parameter	Requirement
PR-01	Maximum power	$5\text{ V} \times 500\text{ mA} = 2.5\text{ W}$
PR-02	Maximum latency	1 s
PR-03	Maximum data rate	300 kbps
PR-04	RAM	6 kB
PR-05	Flash memory	32 kB

2.3 Interface Requirements

Table 3: Interface Requirements

ID	Interface	Requirement
IR-01	Computer $\leftrightarrow$ MCU	USB communication
IR-02	MCU $\leftrightarrow$ LoRa module	SPI communication
IR-03	User $\leftrightarrow$ Computer	Command-line interface (CLI) for entering and receiving messages

2.4 Design Constraints

Table 4: Design Constraints

Constraint	Specification
Budget	\$10 per device
Physical dimensions	70 mm $\times$ 25 mm $\times$ 15 mm
Microcontroller	STM32F042F6P6TR
Radio module	Ra-02 LoRa

3 Preliminary System Architecture

3.1 System Block Diagram

The overall system architecture is shown in Figure 1. The system consists of two computers, two STM32F042-based nodes, and two Ra-02 LoRa modules. The computers communicate with their respective microcontrollers through USB, while the microcontrollers communicate with the LoRa modules through SPI.

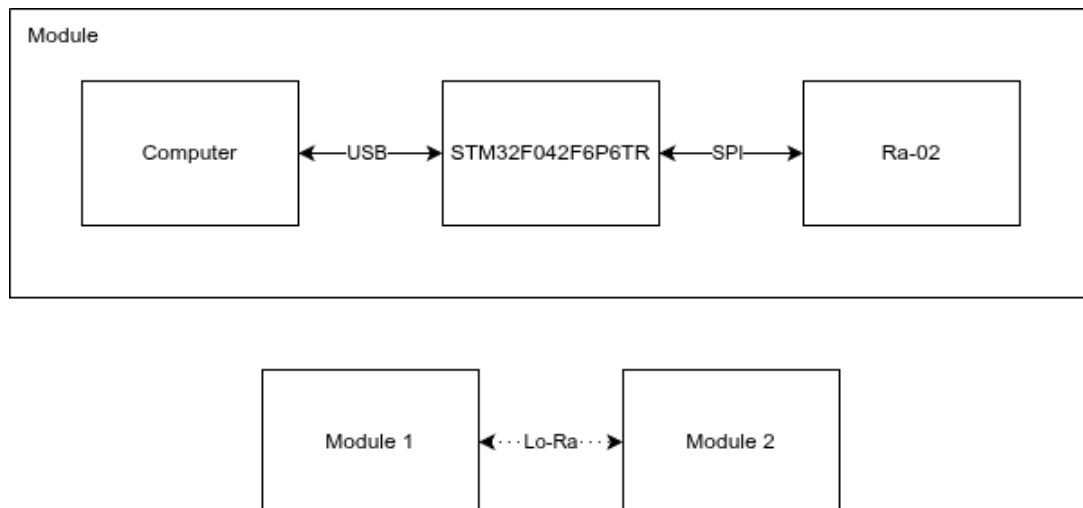


Figure 1: Preliminary system block diagram

3.2 Software Architecture

The software architecture describes the main software modules and the flow of data between the computer, STM32F042 microcontroller, and LoRa module. The software architecture is shown in Figure 2.

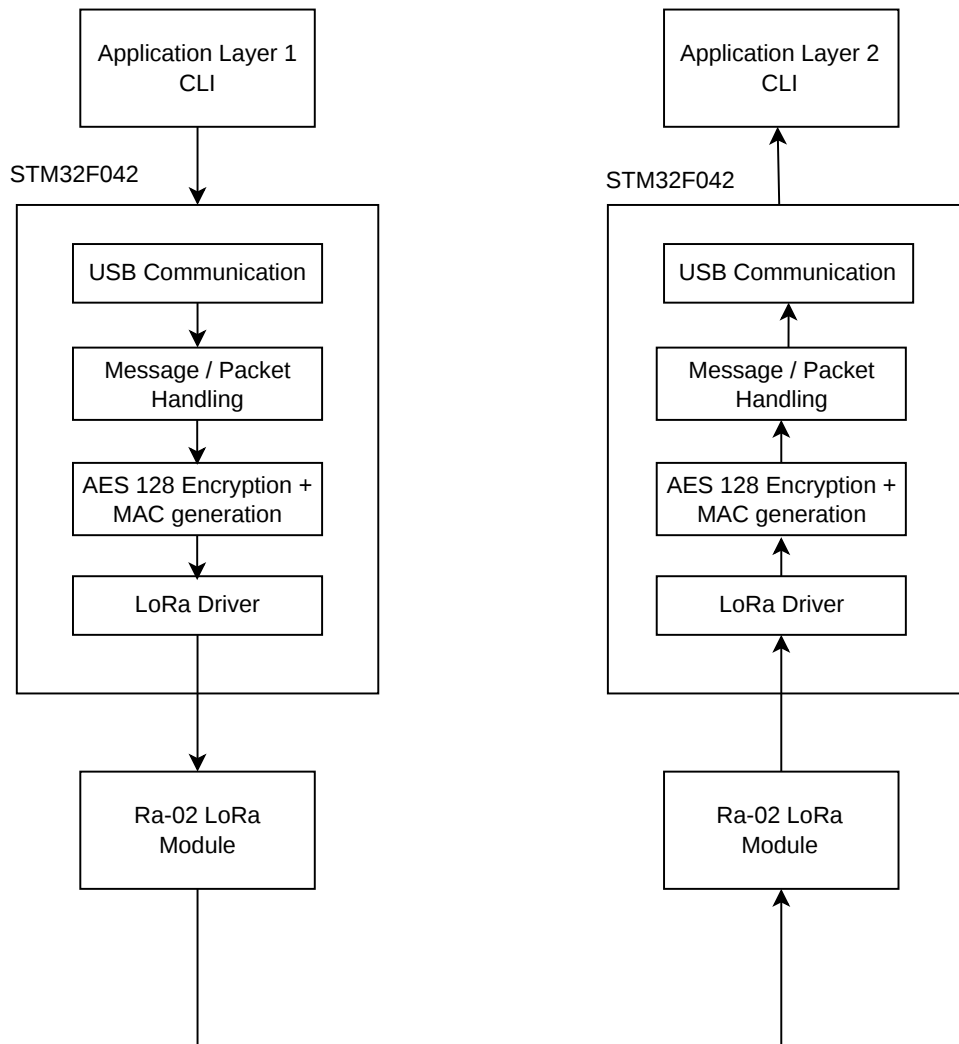


Figure 2: Software architecture and data flow

3.3 Component Selection & Justification

Microcontroller – STM32F042F6P6TR

The STM32F042F6P6TR was selected because it supports the required USB and SPI communication interfaces while providing sufficient processing capability and memory for the intended application. The microcontroller is also supported by the STM32Cube development ecosystem, which simplifies configuration, development, and debugging. Its availability and low cost make it suitable for the project’s low-cost design requirements.

Radio Module – Ra-02

The Ra-02 LoRa module was selected for wireless communication because it is suitable for integration into a prototype board and supports an external antenna through an IPEX connector. Its compact form factor and straightforward hardware integration make it suitable for the proposed embedded communication system.

4 Technical Risk Assessment

The following technical and logistical risks have been identified for the project.

Table 5: Technical and Logistical Risks

Risk	Description	Impact /Likelihood	Mitigation Strategy
Key leakage	If a device is compromised, the pre-shared encryption key may be exposed, compromising the security of the communication.	Medium	Store the key securely on the device and avoid exposing it through the computer-side application or transmitted messages.
Packet loss	A packet may be lost during wireless transmission or reception.	Low	Include a packet identifier in each transmitted packet so that missing packets can be detected and handled appropriately.
Packet collision	Simultaneous or overlapping transmissions may cause packets to interfere with each other.	Medium	Buffer outgoing messages and use controlled transmission and retry mechanisms to reduce the probability of collisions and recover from failed transmissions.

5 Project Management & Task Distribution

5.1 Work Breakdown Structure (WBS)

The project is divided into the following main tasks:

1. USB communication with the computer
2. AES-128 encryption and message authentication
3. SPI communication with the LoRa module
4. CLI application for messaging
5. Final testing and evaluation

5.2 Project Schedule

The planned project schedule is shown in Figure 3. The deadlines are preliminary and subject to change.

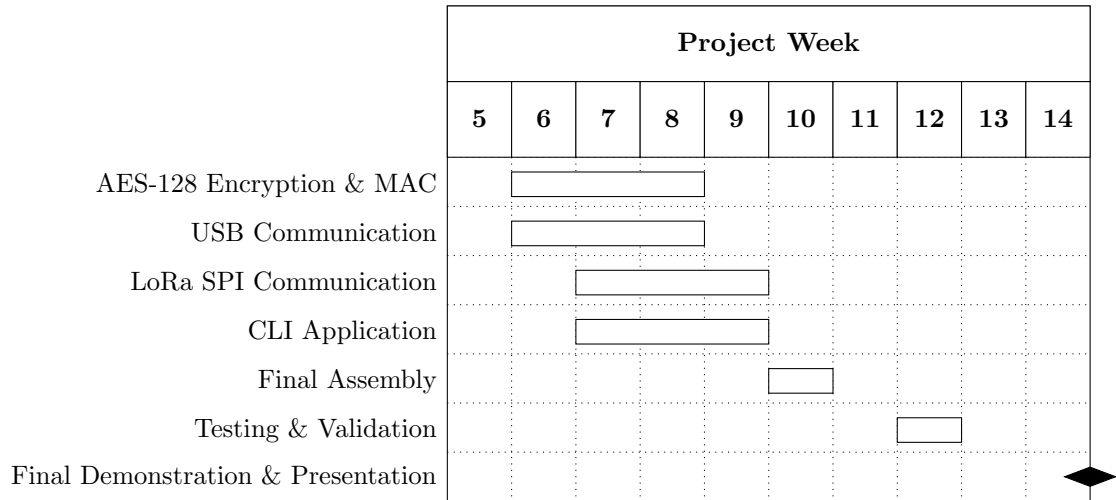


Figure 3: Project schedule and milestones

Note: The deadlines are preliminary and subject to change.

5.3 Team Roles

The responsibilities and deliverables of each team member are defined below.

Table 6: Team Roles and Responsibilities

Team Member	Task	Responsibility	Deliverables
Mher Saribekyan	Hardware Communications	USB and SPI communication, LoRa driver, and packet parser.	STM32 project implementing USB and SPI communication; packet parser for LoRa messages.
Lusine Khachatryan, Anna Grigoryan	Encryption	Implementation of AES-128 encryption and message authentication.	AES-128 encryption and decryption functions and MAC functionality.
Avet Tonoyan	CLI Application	USB serial port communication and user interaction through the command-line interface.	CLI application for sending and receiving data through the USB serial interface.